



EXTENSION 2 MATHEMATICS

2001 TRIAL EXAMINATION

Time : 3 hours + 5 minutes reading time

Instructions:

Attempt all questions

All questions are of equal value

All necessary working should be shown for every question.

Full marks may not be awarded for careless or badly arranged work

A Table of Standard Integrals is provided

Approved calculators may be used

Each question should be answered in a separate booklet

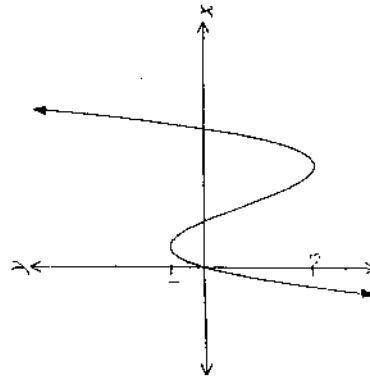
Question 1

- (a) T $(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre O. A line through O, parallel to the tangent at T, meets the ellipse at M and N.

- (i) Show the gradient of the tangent at T is $-\frac{b \cos \theta}{a \sin \theta}$ and find the equation of MN. [3]
- (ii) Show that M and N are $(-a \sin \theta, b \cos \theta)$ and $(a \sin \theta, -b \cos \theta)$. [3]
- (iii) Show that the area of $\triangle TMN$ is independent of θ . [5]
- (b) Describe the locus $|x - 3| + |x + 3| = 10$. [4]

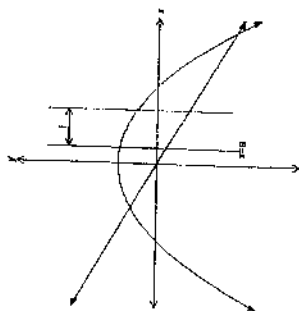
Question 2

- (a)



$y = f(x)$ is drawn above. Copy the diagram into your answer booklet and on the same diagram sketch $y = \log_e f(x)$. [2]

- (d) Consider the area between the curves $y = 3 - x^2$ and $y = -2x$. Suppose that two vertical lines 1 unit apart cross this area.



- (i) If the first line is $x = a$, write an expression for the shaded area. [3]
(ii) Find the maximum value of the shaded area. [2]

Question 4

- (a) Use the substitution $u = x - 1$ to find $\int \frac{x}{\sqrt{x-1}} dx$ [3]
(b) Find the exact value of (i) $\int_1^e \log_e x dx$ [2]
(ii) $\int_0^{\ln 3} e^x \operatorname{cosec}^2(e^x) dx$ [3]
(c) (i) Using the substitution $u = \frac{1}{x}$, show that $\int_0^1 \frac{\ln x}{1+x^2} dx = \int_2^1 \frac{\ln u}{1+u^2} du$ [2]
(ii) Deduce the value of $\int_0^1 \frac{\ln x}{1+x^2} dx$ [2]
(d) Find $\int \frac{\cos x}{\sin x + \sin^2 x} dx$ [3]

- (b) Find the volume of the solid formed when the arc of $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the line $y = 2$ [6]

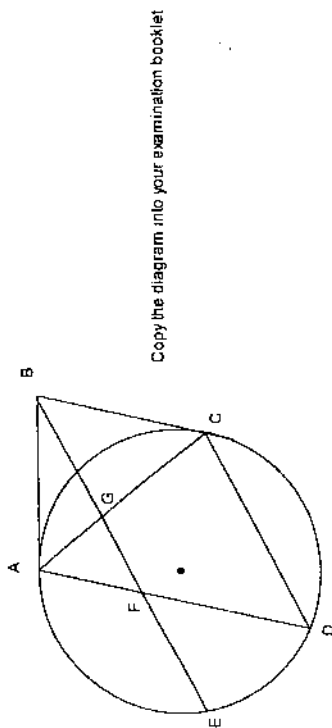
- (c) A dome has a circular base of radius 10 metres. Cross-sections perpendicular to the base and one axis are parabolas whose height is the same as the base width.

- (i) Why would Simpson's rule give the exact area of the parabolic cross-section? [1]
(ii) Show that the area of the parabolic cross-section is $\frac{8y^2}{3}$ square metres. [3]
(iii) Find the volume of the dome. [3]

Question 3

- (a) (i) Express $-1+i$ in modulus argument form [1]
(ii) Hence evaluate $(-1+i)^{10}$ [2]
(b) (i) Find all pairs of integers x and y such that $(x+iy)^2 = -3-4i$ [2]
(ii) Hence or otherwise, solve the quadratic equation $z^2 - 3z + (3+i) = 0$ [2]
(c) Show, by geometrical means or otherwise that, if z_1 and z_2 are complex numbers such that $|z_1| = |z_2|$, then $\frac{z_1 + z_2}{z_1 - z_2}$ is pure imaginary. [3]

(a)



Copy the diagram into your examination booklet

In the diagram EB is parallel to DC. Tangents from B meet the circle at A and C. Prove that

- (i) $\angle BCA = \angle BFA$ [3]
(ii) ABCF is a cyclic quadrilateral [1]
(iii) $DF = CF$ [3]
(b) (i) Draw the graph of $y = \frac{x^4 - 1}{x^2}$ [2]
(ii) On separate axes sketch $y = \tan^{-1}\left(\frac{x^4 - 1}{x^2}\right)$ [2]
(c) (i) On the same axes sketch $y = |x| - 2$ and $y = 4 + 3x - x^2$ [2]
(ii) Hence or otherwise solve $\frac{|x| - 2}{4 + 3x - x^2} > 0$ [2]

(a) Graph the intersection of:

- (a) Graph the intersection of:
 $z\bar{z} \geq 9 \quad z + \bar{z} \leq 8 \quad 0 < \operatorname{Arg}(z) < \frac{\pi}{4}$ [4]
- (b) Let α be the complex root of the polynomial $z^7 = 1$ with the smallest possible argument.
 Let $\theta = \alpha + \alpha^2 + \alpha^4$ and $\delta = \alpha^3 + \alpha^5 + \alpha^6$ [5]
- (i) Explain why $\alpha^7 = 1$ and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$ [1]
- (ii) Show that $\theta + \delta = -1$ and $\theta\delta = 2$ and hence write a quadratic equation whose roots are θ and δ [3]
- (iii) Show that $\theta = -\frac{1}{2} + \frac{i\sqrt{7}}{2}$ and $\delta = -\frac{1}{2} - \frac{i\sqrt{7}}{2}$ [2]
- (iv) Write down α in modulus-argument form, and show that
 $\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$ and $\sin \frac{4\pi}{7} - \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$ [5]

Question 7

- (a) The roots of a cubic equation are α , β and γ , and $\sum \alpha^4 = \alpha^4 + \beta^4 + \gamma^4$. It is given that $\sum \alpha = -1$, $\sum \alpha^3 = 7$, $\sum \alpha^2 = 8$ [2]
- (i) Deduce that the equation is $x^3 + x^2 - 3x - 6 = 0$ [2]
- (ii) Hence evaluate $\sum \alpha^4$ [2]
- (b) (i) If $I_n = \int x(\ln x)^n dx$ for $n \geq 0$, show that $I_n = \frac{1}{2}x^2(\ln x)^n - \frac{n}{2}I_{n-1}$ [2]
- (ii) Hence, find $\int x(\ln x)^2 dx$ [2]
- (c) A particle is projected from the origin at an angle of α° with initial velocity V , and it passes through a point (m, n) .
- (i) Prove that $gm^2 \tan^2 \alpha - 2mV^2 \tan \alpha + gm^2 + 2nV^2 = 0$ [4]
where g is acceleration due to gravity
- (ii) Prove that there are two possible trajectories if $(V^2 - gm)^2 > g^2(m^2 + n^2)$ [3]

Question 8

- (a) A chord AB and a diameter CD, of a circle centre O, intersect at M within the circle. M is not the centre.
- (i) Show that $(CM + MD)^2 > (AM + MB)^2$ [2]
- (ii) Deduce that $(CM - MD)^2 > (AM - MB)^2$ [2]
- (b) A particle of mass m kg falls from rest in a medium where the resistance to motion is mkv when the particle has velocity v m/s.
- (i) Draw a diagram showing the forces acting on the particle. [1]
- (ii) Show that the equation of motion of the particle is $\ddot{x} = k(V - v)$ where V m/s is the terminal velocity of the particle in this medium, and x metres is the distance fallen in t seconds. [2]
- (iii) Find in terms of V and k the time T seconds taken for the particle to attain 50% of its terminal velocity, and the distance fallen in this time. [4]
- (iv) What percentage of its terminal velocity will the particle have attained in time $2T$ seconds? Sketch a graph of v against t showing this information. [3]
- (v) If the particle has reached 87.5% of its terminal velocity in 15 seconds, find the value of k . [1]

End of Examination

Q3d) cont.

$$= \frac{1}{3} [9 - a^3 - 3a^2 - 3a - 1 + 3a^2 + 6a + 3 + a^3 - 3a^2] \quad \text{d) } \frac{\cos x}{\sin x + \sin^2 x} \quad \text{let } u = \sin x$$

$$= \frac{11}{3} - a^2 + a$$

(ii) Max area occurs when $a = \frac{1}{2\sqrt{2}}$

$$\text{max area} = \frac{11}{3} - \frac{1}{4} + \frac{1}{2\sqrt{2}}$$

$$\begin{aligned} \text{4a) } \int \frac{x}{\sqrt{x-1}} dx &= \int \frac{u+1}{u^{1/2}} du \\ &= \int u^{1/2} + u^{-1/2} du \\ &= \frac{2}{3} u^{3/2} + 2u^{1/2} + C \\ &= \frac{2}{3} (x-1)^{3/2} + 2(x-1)^{1/2} + C \end{aligned}$$

$$\begin{aligned} \text{i) } \int_0^e \log x dx & \quad \text{let } u = \ln x \\ &= [x \ln x]_0^e - \int_0^e x \cdot \frac{1}{x} dx \\ &= [e - 0] - [e - 1] \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_0^{\pi/2} e^{\cos x} \cos x dx &= [-\cot e^x]_0^{\pi/2} \\ &= -\cot e^{\pi/2} + \cot e^0 \\ &= \cot 1 - \cot 3 \end{aligned}$$

$$\begin{aligned} \text{i) } \int_0^1 \frac{\ln x}{1+x^2} dx & \quad u = \frac{1}{x}, x = \frac{1}{u} \\ &= \int_1^{\infty} \frac{-\ln u}{1+\frac{1}{u^2}} \cdot \frac{du}{u^2} \\ &= \int_1^{\infty} \frac{-\ln u}{1+u^2} du \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_0^2 \frac{\ln x}{1+x^2} dx &= \int_0.5^1 \frac{\ln x}{1+x^2} dx + \int_1^2 \frac{\ln x}{1+x^2} dx \\ &= \int_0.5^1 \frac{\ln x}{1+x^2} dx - \int_2^1 \frac{\ln x}{1+x^2} dx \\ &= 0 \quad \text{from (i)} \end{aligned}$$

Q4 cont.

$$\frac{d}{dx} \left(\frac{1}{\sin x + \sin^2 x} \right) = \frac{-\cos x (1 + 2 \sin x)}{(\sin x + \sin^2 x)^2}$$

$$= \int \frac{dx}{(\sin x + \sin^2 x)^2}$$

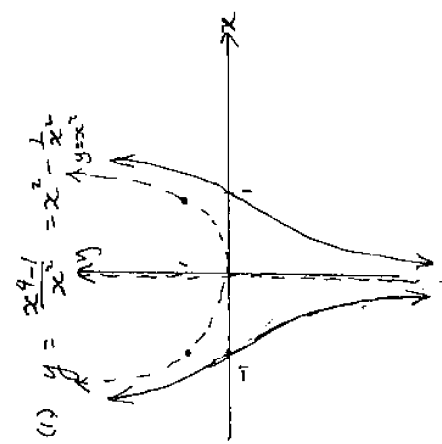
$$\begin{aligned} &= \int \frac{du}{u(u+1)^2} \quad \text{let } u = \sin x \\ &= \int \frac{1}{u} - \frac{1}{u+1} + \frac{1}{(u+1)^2} du \\ &= \ln u - \ln(u+1) - \frac{1}{u+1} + C \end{aligned}$$

Q5a) i) Let $\hat{BCA} = x^\circ$
 $\hat{ABC} = x^\circ$ (L in alt. ang)
 $\hat{BFA} = x^\circ$ (corr. ls, $EB \parallel DC$)
 $\therefore \hat{BCA} = \hat{BFA}$

ii) $\triangle ABC$ is cyclic since $\hat{BCA}, \hat{BFA}$ are equal Ls in same segment.

iii) $AB = BC$ (trans from common pt B)
 $\hat{BAC} = \hat{BCA} = x^\circ$ (base Ls of $\triangle ABC$)
 $\hat{BAC} = \hat{BFC} = x^\circ$ (Ls in same seg. of circle)
 $\therefore \hat{FED} = \hat{FCD} = x^\circ$

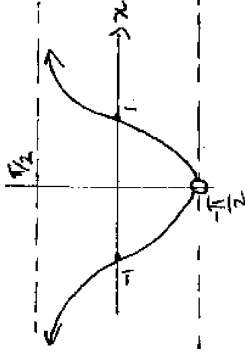
$DF = CF$ (sides opp equal Ls in $\triangle FDC$)



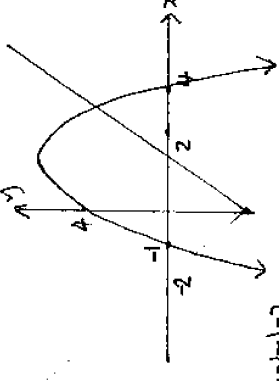
$$\begin{aligned} \text{b) (i) } y &= x^2 - 1 \\ \text{d) } \int_0^1 \frac{\ln x}{1+x^2} dx &= \int_0.5^1 \frac{\ln x}{1+x^2} dx + \int_1^2 \frac{\ln x}{1+x^2} dx \end{aligned}$$

$$\begin{aligned} &= \int_0.5^1 \frac{\ln x}{1+x^2} dx - \int_2^1 \frac{\ln x}{1+x^2} dx \\ &= 0 \quad \text{from (i)} \end{aligned}$$

Q5b) cont.



i)



c) ii)

$$\begin{aligned} y &= x^2 - 2 \\ y &= 4 + 3x - x^2 \\ &= (4-x)(1+x) \end{aligned}$$

$$\frac{|x|-2}{4+3x-x^2} > 0 \quad \text{when}$$

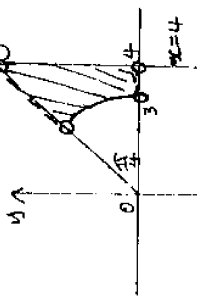
$$|x|-2 > 0 \quad \text{and} \quad 4+3x-x^2 > 0$$

or

$$|x|-2 < 0 \quad \text{and} \quad 4+3x-x^2 < 0$$

$$\text{i.e. } -2 < x < -1 \quad \text{or} \quad 2 < x < 4$$

$$\text{d) } 2\bar{z} \geq 9 \quad \bar{z} + \bar{z} \leq 8 \quad 0 < \arg \bar{z} < \frac{\pi}{4}$$



$$\begin{aligned} 2\bar{z} &= (x+iy)(x-iy) \\ &= x^2 + y^2 \\ \therefore x^2 + y^2 &\geq 9 \end{aligned}$$

Q6 cont.

6

b) α is a root of $z^7 = 1 \quad \therefore \alpha^7 = 1$
 (i) $z^7 - 1 = (z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$
 α is a root so $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$

$$\begin{aligned} \text{ii) } \theta &= \alpha + \alpha^2 + \alpha^4 \quad \delta = \alpha^3 + \alpha^5 + \alpha^6 \\ \theta + \delta &= \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 \\ &= -1 \quad \text{from (i)} \\ \theta \delta &= (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) \\ &= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^7 + \alpha^7 + \alpha^7 + \alpha^7 \\ &= \alpha^4(1 + \alpha + \alpha^2 + 3\alpha^3 + \alpha^4 + \alpha^5 + \alpha^6) \\ &= \alpha^4(0 + 2\alpha^3) \quad \text{from (i)} \\ &= 2\alpha^7 \\ &= 2 \end{aligned}$$

Quadratic is $x^2 + x + 2 = 0$

$$\begin{aligned} \text{iii) } x &= \frac{-1 \pm \sqrt{1-4 \times 2}}{2} \\ &= \frac{-1 \pm \sqrt{-7}}{2} \\ \therefore \theta &= -\frac{1}{2} \pm \frac{\sqrt{-7}}{2} \quad \text{and} \quad \delta = -\frac{1}{2} - \frac{\sqrt{-7}}{2} \end{aligned}$$

$$\begin{aligned} \text{iv) } \alpha &= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \\ \therefore \theta &= \alpha + \alpha^2 + \alpha^4 \\ &= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + i(\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}) \end{aligned}$$

$$\begin{aligned} \text{Equating real and im. parts.} \\ \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} &= -\frac{1}{2} \\ \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \frac{\pi}{7} &= -\frac{1}{2} \\ \text{Also } \cos \frac{8\pi}{7} = -\cos \frac{\pi}{7} \quad \text{and } \sin \frac{8\pi}{7} = -\sin \frac{\pi}{7} \end{aligned}$$

$$\begin{aligned} \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} &= \frac{\sqrt{7}}{2} \\ \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} - \sin \frac{\pi}{7} &= \frac{\sqrt{7}}{2} \end{aligned}$$

Q7a)

i) Consider $x^3 + px^2 + qx + r = 0$

$\sum \alpha = \alpha + \beta + \gamma = -1 \therefore p = 1$

$\sum \alpha^2 = \alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2 \sum \alpha\beta$

$7 = (-1)^2 - 2q$

$q = -3$

$\therefore x^3 + x^2 - 3x + r = 0$

$x^3 = -x^2 + 3x - r$

$\alpha^3 = -\alpha^2 + 3\alpha - r$

$\beta^3 = -\beta^2 + 3\beta - r$

$\gamma^3 = -\gamma^2 + 3\gamma - r$

$\sum \alpha^3 = -(\alpha^2 + \beta^2 + \gamma^2) + 3(\alpha + \beta + \gamma) - 3r$

$8 = -7 + 3(-1) - 3r$

$r = -6$

$\therefore$ cubic is $x^3 + x^2 - 3x - 6 = 0$

ii) $x^4 + x^3 - 3x^2 - 6x = 0$

$x^4 = -x^3 + 3x^2 + 6x$

$\alpha^4 = -\alpha^3 + 3\alpha^2 + 6\alpha$ etc.

$\therefore \sum \alpha^4 = -(\alpha^3 + \beta^3 + \gamma^3) + 3(\alpha^2 + \beta^2 + \gamma^2) + 6(\alpha + \beta + \gamma)$

$= -8 + 3 \times 7 + 6 \times 1$

$= 7$

b) i) $I_n = \int x (\ln x)^n dx$ $n \geq 0$

$= uv - \int v du$

$u = (\ln x)^n$

$\frac{du}{dx} = n (\ln x)^{n-1} \cdot \frac{1}{x}$

$\frac{dv}{dx} = x$

$v = \frac{x^2}{2}$

$I_n = \frac{x^2}{2} (\ln x)^n - \int \frac{x^2}{2} n (\ln x)^{n-1} \cdot \frac{1}{x} dx$

$= \frac{x^2}{2} (\ln x)^n - \frac{n}{2} \int x (\ln x)^{n-1} dx$

$= \frac{x^2}{2} (\ln x)^n - \frac{n}{2} I_{n-1}$

Q7b) cont.

ii) $\int x (\ln x)^2 dx = \frac{x^2}{2} (\ln x)^2 - \frac{x^2}{2} \int x \ln x dx$

$= \frac{x^2}{2} (\ln x)^2 - \frac{x^2}{2} (\ln x) + \frac{1}{2} \int x (\ln x)^0 dx$

$= \frac{x^2}{2} (\ln x)^2 - \frac{x^2}{2} \ln x + \frac{x^2}{4} + C$

c) $\begin{matrix} t=0 \\ \dot{x} = v \cos \alpha \\ \dot{y} = v \sin \alpha \\ x=0 \\ y=0 \end{matrix}$

$\ddot{x} = 0$

$\ddot{y} = -g$

$\dot{x} = v \cos \alpha$

$\dot{y} = -gt + v \sin \alpha$

$x = vt \cos \alpha$

$y = -\frac{1}{2} gt^2 + v t \sin \alpha$

$\Rightarrow \textcircled{B}$

$y = -\frac{1}{2} g \left(\frac{x}{v \cos \alpha} \right)^2 + v \left(\frac{x}{v \cos \alpha} \right) \sin \alpha$

$= -\frac{1}{2} g \frac{x^2}{v^2 \cos^2 \alpha} + x \tan \alpha$

Path passes through (m, n)

$\therefore n = -\frac{1}{2} g \frac{m^2}{v^2} (1 + \tan^2 \alpha) + m \tan \alpha$

$2V^2 n = -gm^2 \tan^2 \alpha + 2V^2 m \tan \alpha$

$gm^2 \tan^2 \alpha - 2V^2 m \tan \alpha + 2nV^2 = 0$

ii) Two possible trajectories if this quadratic in $\tan \alpha$ has 2 roots

ie if $\Delta > 0$

$(-2V^2 m)^2 - 4gm^2 (gm^2 + 2nV^2) > 0$

$4m^2 V^4 - 8gm^2 nV^2 - 4gm^4 > 0$

$4m^2 > 0$ so divide by $4m^2$

$V^4 - 2gnV^2 - g^2 m^2 > 0$

$V^4 - 2gnV^2 + (gm)^2 > g^2 m^2 + gn^2$ (completing)

$(V^2 - gn)^2 > g^2 (m^2 + n^2)$

Q8a) i) $AP (CM+MD)^2 > (AM+MB)^2$

(i) $CD > AB$ (diagonals > chord)

$CM+MD > AM+MB$

$(CM+MD)^2 > (AM+MB)^2$

ii) i) $AP (CM-MD)^2 > (AM-MB)^2$

from (i)

$(CM+MD)^2 > (AM+MB)^2$

$\therefore (CM-MD)^2 + 4CM \cdot MD > (AM-MB)^2 + 4AM \cdot MB$

Now $CM \cdot MD = AM \cdot MB$ (Intersecting chords)

$\therefore (CM-MD)^2 > (AM-MB)^2$

b) i) $\downarrow mg \uparrow mv$

ii) $ma = mg - mv$

$\ddot{x} = g - kv$

$= k(g/k - v)$

$\ddot{x} \rightarrow 0$ as $v \rightarrow g/k$

$\therefore V = g/k$ is the terminal vel.

$\therefore \ddot{x} = k(V - v)$

iii) $\frac{dv}{dt} = k(V - v)$

$\therefore \frac{dv}{V - v} = \frac{k}{V} dt$

$x = \frac{1}{k} \int \frac{1}{V - v} dv$

$= -\frac{1}{k} \ln(V - v) + C$

when $t=0, v=0$

$\therefore C = \frac{1}{k} \ln V$

$x = \frac{1}{k} \ln \frac{V}{V - v}$ (1)

when $v = 50\% V = 0.5V$

$x = \frac{1}{k} \ln \left(\frac{V}{0.5V} \right)$

$= \frac{1}{k} \ln 2$

Distance: $kt = \ln \frac{V}{V - v}$

$e^{kt} = \frac{V}{V - v}$

$V - v = Ve^{-kt}$

$v = V(1 - e^{-kt})$

$\frac{dx}{dt} = V(1 - e^{-kt})$

$x = V \int (1 - e^{-kt}) dt$

$= V \left[t + \frac{1}{k} e^{-kt} \right] + C$

when $t=0, x=0 \therefore C = -\frac{V}{k}$

$x = V \left[t + \frac{1}{k} e^{-kt} \right] - \frac{V}{k}$

when $t = \frac{1}{k} \ln 2$

$x = V \left[\frac{1}{k} \ln 2 + \frac{1}{k} e^{-\ln 2} \right] - \frac{V}{k}$

$= \frac{V}{k} \left[\ln 2 - \frac{1}{2} \right]$

$= \frac{V}{k} [2 \ln 2 - 1]$

From (i) $kt = \ln \frac{V}{V - v}$

$e^{kt} = \frac{V}{V - v}$

$v = V(1 - e^{-kt})$

Now $T = \frac{1}{k} \ln 2$

$2T = \frac{1}{k} \ln 2$

$\therefore v = V(1 - e^{-k \cdot \frac{1}{k} \ln 2})$

$= V(1 - e^{-\ln 2})$

$= V(1 - \frac{1}{2})$

$= 75\% V$

$\therefore$ At time $2T$, velocity is 75% of V

[Note: Exponential so $(V - v)$ halves every T

When $t=15, v=0.875V$

$0.875V = V(1 - e^{-15k})$

$e^{-15k} = 1 - 0.875 = 0.125$

$-15k = \ln 0.125$

$k = -\frac{1}{15} \ln 0.125$

(≈ 0.1386)